

When Yield Curves Invert Together: Predicting Currency Carry Losses

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Abstract

Currencies face country-specific and global risks. An inversion of one country's yield curve may reflect either, but synchronized inversions across several non-U.S. G10 economies are more likely to contain information about global conditions. I combine these inversions into a state variable that predicts unusually poor carry-trade spot returns. Annualized carry-trade spot returns are 12.9 percentage points lower when the lagged state is active than when inactive, with inference based on fifteen episodes. High-interest-rate currencies and currencies with greater predetermined equity-market betas experience larger losses, supporting the interpretation that the state contains information about global rather than country-specific risk.

JEL: E43, F31, G12, G15.

Keywords: currency carry trade, yield-curve inversion, exchange-rate predictability, global risk, G10 currencies.

1 Introduction

Currencies face both country-specific and global risks. This distinction is especially important for the carry trade. Diversification removes part of a currency's idiosyncratic risk, but it does not protect a portfolio that is long several high-interest-rate currencies when those currencies depreciate together. Because interest income accrues gradually, one shared exchange-rate movement can erase many months of carry. The empirical question is whether bond markets reveal such a state before the currency loss occurs.

An inversion of one country's yield curve is ambiguous. It may reflect domestic weakness, disinflation after a national tightening cycle, or a change in term premia. Synchronized inversions across several non-U.S. G10 economies are more likely to contain information about a global disturbance, although synchronization alone cannot distinguish global growth news from common policy or term-premium shocks. I combine fresh 10-year/2-year inversions across nine non-dollar G10 economies into a monthly state and ask whether that state predicts currency returns.

The construction is deliberately path dependent. A curve enters when it crosses into inversion and remains live until its slope has increased for two consecutive months. The synchronized-inversion state is active when at least two curves are live. Fresh entry separates new crossings from stale inversions; confirmed release keeps the state active when curves begin to re-steepen, including periods in which expected easing is being delivered. The state observed at the end of month t predicts the return during month $t + 1$.

The 1988–2026 sample contains 92 active months in fifteen episodes. Annualized carry-trade spot returns are 12.9 percentage points lower when the lagged state is active: 2.9 percent outside the state and -10.0 percent inside it. Total carry returns fall from 6.5 to -4.9 percent. The result is not only a label for the largest crashes. After removing each return series' full-sample bottom-decile months, the spot-return gap remains -7.2 percentage points per year. The center of the distribution also shifts: the monthly carry-spot median falls from 0.35 to -0.76 percent.

The cross-section gives the result economic content. Carry buys high-interest-rate currencies and funds them with low-interest-rate currencies. Those rate rankings are correlated with exposure to global markets: the rank correlation between currencies' average interest differentials and their average predetermined equity betas is 0.72. In active-state months, high-rate currencies depreciate relative to low-rate currencies,

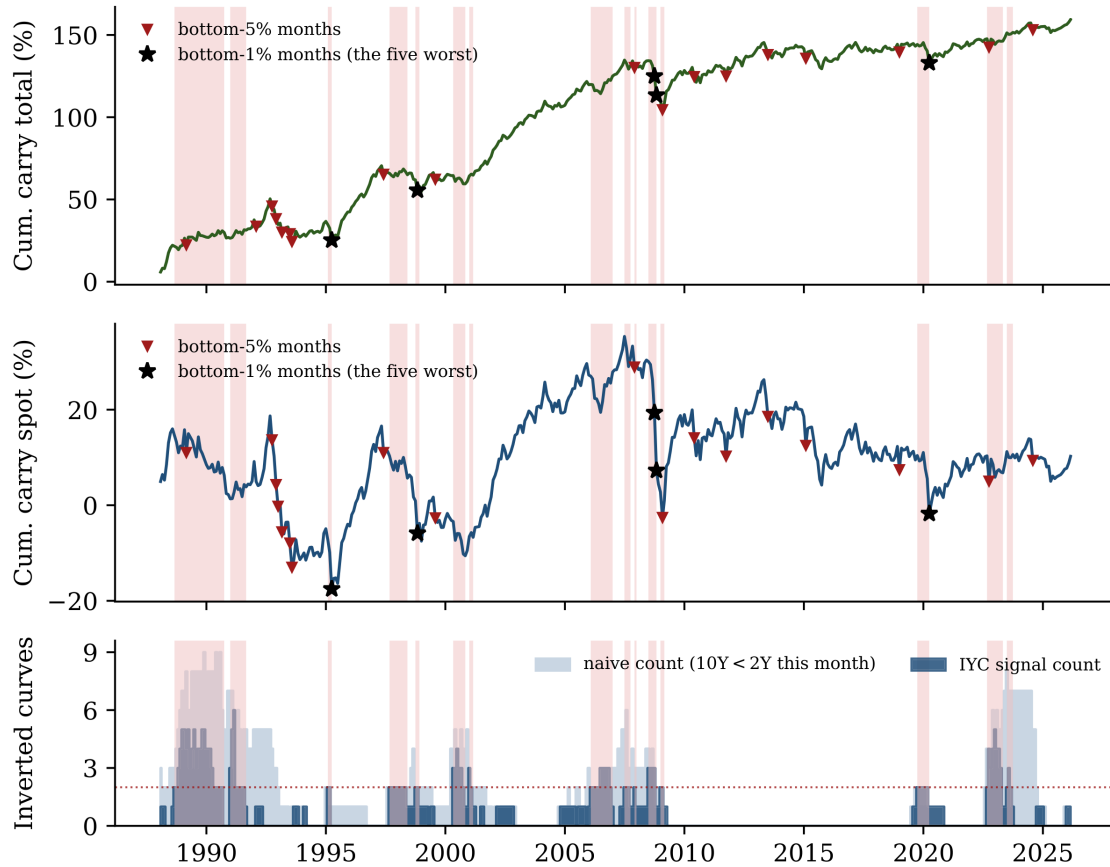


Figure 1: Carry returns and synchronized inversions. The light line counts curves currently inverted; the dark line counts fresh inversions that remain live under the release rule. The dotted line is the threshold of two. Shading denotes the end-of-month synchronized-inversion state, used with a one-month lag in every return regression. Tail markers show that the largest losses follow information already visible in the state, including episodes in which the current-inversion count has declined.

and bilateral losses become larger as predetermined equity beta rises. Across nine currencies, the beta gradient explains 77 percent of the variation in estimated state coefficients; a portfolio formed directly on expanding-window equity betas loses 7.5 percentage points per year in spot returns in the state. The panel interaction has the expected negative sign but weak episode-level inference. Taken together, the estimates support an exposure ordering, not a precisely identified structural beta.

This ordering links two established literatures. National yield slopes forecast activity and reflect both expected short-rate paths and term premia (Harvey, 1988; Estrella and Hardouvelis, 1991; Estrella and Mishkin, 1998; Campbell and Shiller, 1991; Hamilton and Kim, 2002). International term structures also contain a shared component (Plosser and Rouwenhorst, 1994; Bernard and Gerlach, 1998; Diebold et al., 2008; Chinn and Kucko, 2015). The carry literature, meanwhile, shows that high-interest-rate currencies earn average premia but suffer negatively skewed losses in common stress states (Fama, 1984; Brunnermeier et al., 2008; Lustig et al., 2011; Menkhoff et al., 2012; Farhi and Gabaix, 2016). The contribution here is to shift the unit of observation from one country’s slope or a smooth global level factor to a cross-country configuration of fresh threshold crossings dated before the currency payoff.

The exposure ordering also clarifies why interest rates matter twice. The differential is the carry trade’s known income, but it also ranks the currencies that bear the largest state-contingent spot losses. Commodity exposure, country size, trade-network position, and intermediary demand can all align high rates with sensitivity to global conditions (Hassan, 2013; Ready et al., 2017; Richmond, 2019; Gabaix and Maggiori, 2015). The paper does not distinguish those channels. It shows that the loss is concentrated in currencies for which a predetermined market measure indicates greater exposure, rather than appearing as a uniform dollar movement.

Several results delimit this interpretation. One live inversion has the opposite return sign from two; the U.S. inversion alone has a small and imprecise coefficient; and the continuous average G10 slope, lagged financial conditions, and lagged VIX do not span the synchronized state. The five worst monthly carry returns occur under the lagged state, yet trimmed estimates and calendar splits show that 2008 and 2020 are not the entire result. These comparisons distinguish synchronization from nearby curve and risk measures. They do not identify the shock that synchronizes curves, and they do not remove the central design risk: the tenor, threshold, and release rule were assessed on the same historical sample.

Finally, a predictable negative conditional return is not an automatic implication of disaster exposure. In a frictionless benchmark, publicly known bad-state exposure should command compensation. The paper therefore treats slowly adjusting compensation as one candidate interpretation: fresh bond-market information can raise perceived loss risk before currency compensation fully adjusts. This mechanism explains why entry and timing may matter, but the return evidence does not identify the friction. The durable contribution is narrower and more testable: synchronized foreign yield-curve inversions predict where common currency losses subsequently occur.

Section 2 defines returns, the state, and timing. Sections 3–4 present the design and evidence. Section 5 develops the exposure ordering and economic interpretation; Section 6 states the main limits. The online appendix gives the exact algorithm, derivations, supplementary estimates, and public-data checks.

2 Measurement and Timing

2.1 Currency returns and the shared-loss problem

Consider a U.S. investor holding currency i . Let $s_{i,t}$ be the log dollar price of one unit of foreign currency, so $\Delta s_{i,t+1} > 0$ is foreign-currency appreciation. Let $d_{i,t} \equiv (i_{i,t} - i_{US,t})/12$ be the one-month interest differential, formed from annualized money-market rates. The excess-return proxy is

$$rx_{i,t+1} = d_{i,t} + \Delta s_{i,t+1}. \quad (1)$$

The differential is known at t ; the spot change is realized from t to $t + 1$. Under covered interest parity this sum corresponds to a forward return apart from basis and measurement differences (Du et al., 2018). Because the paper uses money-market differentials rather than executable forward quotes, the object is an asset-pricing return proxy, not a transaction-cost-adjusted strategy.

Each month, the carry portfolio buys the three highest-rate currencies and sells the three lowest-rate currencies among the nine non-dollar G10 currencies. If $\mathcal{H}_t$ and $\mathcal{F}_t$ denote the high-rate and funding sets, then

$$\begin{aligned} rx_{t+1}^{\text{carry}} &= \frac{1}{3} \sum_{i \in \mathcal{H}_t} rx_{i,t+1} - \frac{1}{3} \sum_{i \in \mathcal{F}_t} rx_{i,t+1} \\ &\equiv C_t + X_{t+1}. \end{aligned} \quad (2)$$

Here C_t is the long-minus-short interest differential and X_{t+1} is the corresponding spot return. The sample-average income leg is 3.9 percent per year and is positive in every month, with a minimum of 0.7 percent. Carry therefore loses only when the spot leg is more negative than the income leg is positive.

To organize the cross-sectional test, write the schematic decomposition

$$\Delta s_{i,t+1} = -\lambda_i g_{t+1} + e_{i,t+1}, \quad (3)$$

where g_{t+1} is an adverse return component shared across currencies, λ_i is currency i 's latent exposure, and $e_{i,t+1}$ is idiosyncratic. Diversification reduces the idiosyncratic component but not the shared component when the two portfolio legs have different average exposures. Here “shared” is statistical; “global” is the economic interpretation.

Equation (3) is an organizing device, not an estimated structural model.

2.2 The synchronized-inversion state

Let $q_{i,t} \equiv y_{i,t}^{10Y} - y_{i,t}^{2Y}$ denote country i 's end-of-month yield slope. An inversion can reflect expected easing, term-premium variation, or both (Campbell and Shiller, 1991; Hamilton and Kim, 2002; Bauer and Mertens, 2018). Several fresh crossings are more suggestive of a shared disturbance than one crossing, although regional shocks and coordinated policy can also synchronize curves.

Define $I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}$. A curve enters the live set when it freshly crosses from $I_{i,t-1} = 0$ to $I_{i,t} = 1$. It remains live until $q_{i,t}$ has increased for two consecutive months. A released curve cannot re-enter while still inverted; it must first become nonnegative and then cross below zero again. Let $L_{i,t}$ be the resulting live status. The central state is

$$N_t = \sum_{i=1}^9 L_{i,t}, \quad S_t = \mathbf{1}\{N_t \geq 2\}. \quad (4)$$

Thus $L_{i,t}$ is path dependent, whereas $I_{i,t}$ depends only on the current slope. The state measured at the end of month t conditions the return during month $t + 1$.

The live rule has two economic roles. Fresh entry excludes inversions already embedded in prices for many months. Confirmed release allows the state to remain active as curves re-steepen, which can coincide with delivered easing and delayed currency adjustment. In the sample, a raw two-or-more current-inversion state is active in 165 months, compared with 92 for the live state, yet switches off before March 1995, October 2008, and March 2020—the three largest carry-loss months. This historical fit is informative and also creates specification-search risk. Online Appendix A states the transition rules and boundary cases exactly.

2.3 Data, information set, and predetermined exposure

The monthly sample runs from January 1988 through February 2026. The currency universe comprises the Australian dollar, Canadian dollar, Swiss franc, euro (Deutsche mark before 1999), British pound, Japanese yen, Norwegian krone, New Zealand dollar, and Swedish krona. The dollar is the numeraire and does not enter either the sort or the baseline state count. The U.S. curve is retained as a benchmark.

State incidence varies across monetary regimes. It is active in 33 of 84 months during

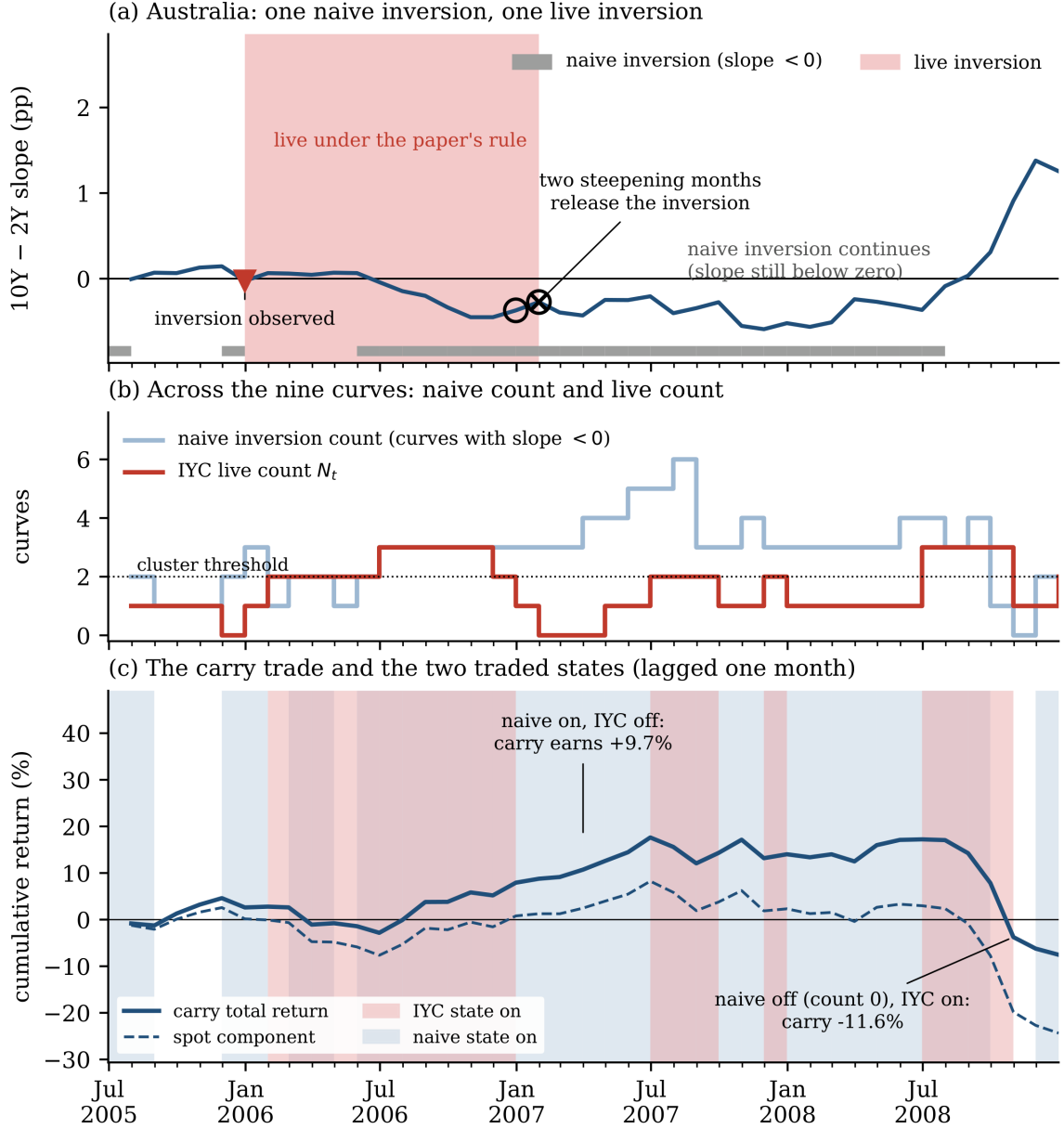


Figure 2: Live-state construction, 2005:07–2008:12. Panel (a) shows the Australian 10Y–2Y slope and its live status. Panel (b) compares the current-inversion and live counts. Panel (c) plots cumulative carry returns under the two lagged states. Fresh entry removes stale inversions; confirmed release can retain the state after the current-inversion count declines.

1988–94 but only 3 of 120 months during 2010–19, when many policy rates were near their lower bounds. The state is therefore not a continuously available global factor; it appears mainly when international rate cycles create enough cross-country slope variation. Online Appendix A reports the full era counts.

The baseline state uses end-of-month 2-year and 10-year government yields. Portfolio weights use one-month money-market rates known at t . Return regressions use no information dated after t to predict month $t + 1$. This convention prevents mechanical look-ahead but does not eliminate asynchronous market closes, stale quotes, or later data revisions.

The exposure test estimates a currency’s equity beta on an expanding window:

$$rx_{i,\tau} = a_i + \hat{\beta}_{i,t} r_\tau^M + u_{i,\tau}, \quad \tau \leq t - 1. \quad (5)$$

The estimate $\hat{\beta}_{i,t}$ is available before the return from t to $t + 1$ and requires at least 36 prior monthly observations. The one-month buffer follows the original construction: the estimate paired with S_t uses currency and equity returns only through $t - 1$. The beta is an empirical proxy for the latent exposure λ_i in equation (3); it need not equal that primitive. S&P 500 total returns form the market measure.

Table 1: Notation and timing

Symbol	Definition	Timing
$t; t + 1$	Information month-end; subsequent holding month	Information $\rightarrow$ return
$s_{i,t}; \Delta s_{i,t+1}$	Log dollar price of currency i ; its one-month change	Spot change realized in $t + 1$
$d_{i,t}$	Monthly foreign-minus-U.S. interest differential	Known at t
$rx_{i,t+1}$	Currency excess-return proxy	Realized in $t + 1$
$C_t; X_{t+1}$	Carry income leg; carry spot leg	Known at t ; realized in $t + 1$
$q_{i,t}$	Country i ’s 10Y–2Y yield slope	Observed at month-end t
$I_{i,t}; L_{i,t}$	Current inversion; path-dependent live inversion	Information at t
$N_t; S_t$	Live-curve count; synchronized-inversion state	Predict $t + 1$
$\lambda_i; \hat{\beta}_{i,t}$	Latent shared-factor exposure; estimated equity-beta proxy	Beta uses data through $t - 1$
W_t	Vector of lagged benchmarks or controls	Known at t

Notes: Equations use the information-time convention $t \rightarrow t + 1$. Tables indexed by the return month equivalently report S_{t-1} .

3 Empirical Design

3.1 Predictive regression and inference

All return tests use one timing convention: information at t predicts the return at $t + 1$. The general specification is

$$Y_{t+1} = \alpha + bS_t + \Gamma'W_t + \varepsilon_{t+1}, \quad (6)$$

where W_t contains the relevant lagged benchmarks or controls. The headline conditional-mean difference sets W_t empty. With a binary state, b is the active-minus-inactive mean difference; monthly return coefficients are multiplied by twelve for reporting. In tables indexed by return month, equation (6) is equivalently written with S_{t-1} .

The state’s 92 active months form fifteen contiguous episodes, fourteen longer than one month. Treating all active months as independent would overstate the information in the sample. I report Newey–West statistics with 12 lags and, for binary-state tests, a wild episode bootstrap that assigns Rademacher weights to contiguous episodes and imposes the null. The episode bootstrap is the headline inference because continuation months do not create independent synchronized shocks. Fifteen episodes nevertheless provide limited common-state variation; no resampling method can manufacture a missing holdout.

Distributional diagnostics report medians, lower quantiles, skewness, expected shortfall, and conditional variance. A separate diagnostic removes each outcome’s full-sample bottom-decile observations from both states before re-estimating equation (6). Because that deletion conditions on realized returns, it is not a preferred estimator. It answers only whether a handful of extreme observations mechanically generate the mean gap.

3.2 Benchmarks and controls

The benchmark hierarchy moves from nearby measurement alternatives to broader risk states. The raw current-inversion state asks whether fresh entry and confirmed release add information. The U.S. inversion asks whether the conventional single-country proxy spans the international state. The average G10 slope and its change ask whether the threshold is merely a nonlinear transformation of a smooth global slope. Nine country-specific live states ask whether synchronized crossings add information beyond

their individual components.

Lagged NFCI and VIX controls address predictive spanning: do variables already known at t absorb the state’s forecast? Contemporaneous dollar and currency-volatility factors answer a different question: do those realized factors account for the loss after it occurs? Neither design identifies the shock represented by S_t . Joint specifications are reported without sequential orthogonalization because the ordering of closely related yield-curve transformations would otherwise determine which variable receives their shared variation.

3.3 Exposure ordering

The bilateral state coefficients come from

$$\Delta s_{i,t+1} = a_i + b_i S_t + e_{i,t+1}. \quad (7)$$

The nine estimates $\hat{b}_i$ are compared with currencies’ average predetermined equity betas. The slope, fitted intercept at zero beta, and uncertainty describe the exposure gradient. A beta-sorted portfolio and a panel interaction $S_t \times \hat{\beta}_{i,t}$ provide complementary tests: the portfolio is economically transparent; the panel uses monthly variation but still has only fifteen common-state episodes.

This design separates three claims. Prediction asks whether S_t forecasts returns. Exposure accounting asks whether the realized losses are ordered by a predetermined market characteristic. Mechanism identification would require independent variation that distinguishes global growth, policy, term-premium, inflation, and intermediary channels. The first two are directly estimated; the third is not.

3.4 Failure patterns

Four findings would materially weaken the paper: the return gap disappearing after tail removal; one-country or smooth-slope benchmarks spanning the state; no relation between currency losses and predetermined exposure; or a real-time implementation requiring future observations. The reported estimates run counter to these patterns, but all variants were examined on the same history. A prospective frozen-rule evaluation remains the decisive external test.

4 Main Evidence

4.1 A large spot-return shift

Annualized carry-trade spot returns are 12.85 percentage points lower when the lagged synchronized-inversion state is active, with inference based on fifteen episodes. Spot returns average 2.85 percent in the 366 inactive months and -10.00 percent in the 92 active months. Total carry returns fall from 6.45 to -4.86 percent, a difference of -11.31 percentage points. Episode-bootstrap p -values are 0.001 and 0.003.

Table 2: The carry trade in and out of the IYC signal

Portfolio component	Annualized mean		State regression	
	No signal	Signal	Difference	Episode wild p
Carry spot	2.85	-10.00	-12.85	0.001
Carry total	6.45	-4.86	-11.31	0.003
High-beta spot	1.26	-6.27	-7.53	0.010
High-beta total	3.18	-3.64	-6.82	0.013

Notes: Returns are annualized percentages. The no-signal state contains 366 months and the signal state 92 months over 1988:01–2026:02. The difference is the coefficient on the one-month-lagged IYC-state indicator. Wild-bootstrap p -values cluster by the state’s fifteen contiguous episodes ($B = 9,999$).

The accounting matters. Carry income averages 3.9 percent per year and remains positive in every month, yet it cushions only part of the active-state depreciation. The result is therefore not low carry income; it is a spot movement large enough to overwhelm known income. The strategy’s annualized Sharpe ratio falls from 0.89 outside the state to -0.51 inside it.

The distribution shifts beyond the mean. The monthly spot median falls from 0.35 percent outside the state to -0.76 percent inside; the 25th percentile moves from -0.93 to -1.98 percent, and the 10th percentile from -2.35 to -3.94 percent. Spot-return expected shortfall worsens from -4.80 to -8.14 percent. By contrast, the conditional variance ratio is 1.74 with permutation $p = 0.14$. Location and tail frequency shift, while the increase in variance is not statistically detectable.

Removing each return series’ 46 bottom-decile months leaves a carry-spot coefficient of -7.24 percentage points per year ($p = 0.006$) and a total-return coefficient of -5.50 ($p = 0.024$). This outcome-selected diagnostic does not estimate a counterfactual without crashes. It shows that the mean difference is not arithmetically carried by only

the most extreme observations.

4.2 Why synchronization matters

The count displays a striking sign change. In 88 months with exactly one live inversion, carry spot and total returns are 8.33 and 8.78 percentage points per year higher than in the remainder of the sample, with episode-bootstrap p -values of 0.005 and 0.003. At exactly two live inversions, the coefficients reverse to -15.72 and -15.13 percentage points, each with $p \leq 0.003$. This contrast supports the economic distinction between one-country and synchronized information within the sample. It does not identify two as a universal cutoff.

Nearby bond-market states do not reproduce the result. The U.S. 10-year/2-year inversion is active in 57 lagged months and has carry-spot and total-return coefficients of -1.90 ($p = 0.484$) and -1.34 ($p = 0.637$). Adding the average G10 slope, its one-month change, and the U.S. inversion leaves the synchronized-state coefficient at -13.45 for spot returns ($p = 0.003$) and -12.44 for total returns ($p = 0.012$). The continuous slope variables are imprecise.

Adding all nine country-level live states makes the synchronized coefficient more negative. That increase is not evidence of a cleaner causal effect: the regressors are related transformations of a small set of curves, and individual states absorb many benign one-country episodes. The defensible conclusion is that neither a smooth global slope nor the individual curve states span the cross-country configuration.

4.3 Risk controls, timing, and stability

Lagged observed risk conditions do not absorb the state. With the NFCI level and change, the carry-spot coefficient is -12.84 percentage points ($p < 0.001$). Adding VIX level and change in the post-1990 sample gives -13.18 ($p = 0.004$). Contemporaneous dollar and global-FX-volatility factors reduce the coefficient only to -11.5 ($p = 0.002$). The lagged controls address forecast spanning; the contemporaneous controls show that the portfolio loss is not merely the average dollar movement or the particular volatility innovation in those regressions.

A U.S. term-premium adjustment gives a more mixed result than the simple controls. After stripping each national slope's fitted exposure to the U.S. ACM slope premium, 83 percent of inversions remain. The high-beta state effect remains negative and precise

(-6.2 percentage points, $p = 0.004$), whereas the carry effect roughly halves to -5.4 and is imprecise ($p = 0.11$). Thus a common U.S. term-premium component does not explain the exposure result, but it may account for part of the carry estimate; it is not a decomposition of foreign-country term premia.

The chronology is genuinely predictive under the stated month-end convention. All five worst monthly carry returns occur under the lagged state. March 1995, October 2008, and March 2020 arrive after the raw current-inversion count has receded but while the previously observed live state still conditions the return. Applying the rule recursively produces no later revisions to prior classifications. One additional month of lag roughly halves the carry coefficient, consistent with short-lived information rather than a permanent regime.

Calendar and episode checks are supportive but not external validation. The carry-spot coefficient is -11.41 percentage points before 2005 and -15.85 afterward. Leave-one-episode-out carry estimates remain within roughly one-third of the full-sample estimate. The beta-sorted portfolio is less stable across halves, with its adverse association concentrated after 2005. Online Appendix C reports these results and the rule grids in full.

The main evidence therefore supports an economically large within-sample predictive association: synchronized fresh inversions forecast lower subsequent carry spot returns, and the result is not reproduced by the most obvious single-curve, smooth-slope, or lagged-risk alternatives. The next section asks why the losses fall disproportionately on the high-rate currencies that carry holds.

5 Where the Losses Land

5.1 Rates and predetermined equity exposure

The carry result already contains one cross-sectional ordering: the portfolio loses because the high-rate currencies it buys depreciate relative to the low-rate funding currencies. The bilateral estimates in equation (7) provide a second ordering. State coefficients become more negative as currencies' predetermined equity-market betas rise.

Australia and New Zealand have average betas of 0.20 and 0.13 and annualized state spot coefficients of -13.21 and -13.28 percentage points. Japan and Switzerland have

betas near zero and imprecise coefficients of 2.35 and -0.89 . Across all nine currencies, regressing the state coefficient on average beta gives a slope near -60 percentage points per beta unit, an intercept of -0.75 , and $R^2 = 0.77$. With nine observations and no displayed confidence region for the fitted intercept, this is descriptive evidence of a gradient; it does not show that all global exposure vanishes at zero equity beta.

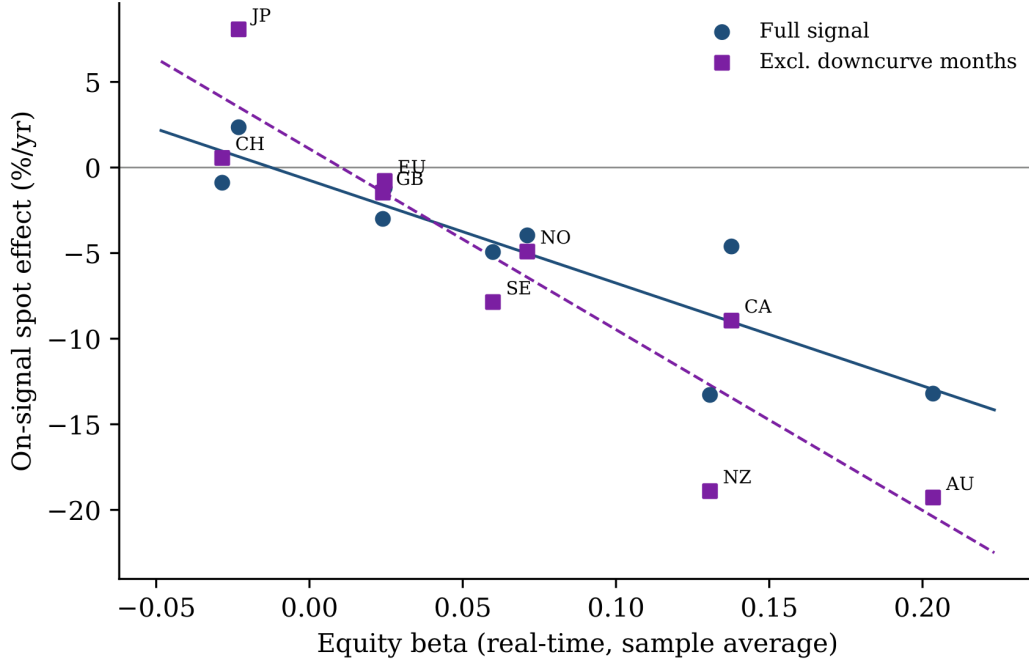


Figure 3: State-dependent spot-return coefficients and predetermined equity beta, one point per currency. Circles and the solid fit show the baseline state (slope -60 , $R^2 = 0.77$). Squares and the dashed fit exclude exploratory downcurve months. The baseline inference uses the solid fit.

Table 3: How active-state losses align with exposure

Object	Estimate	Episode wild p	Interpretation
Rate–beta rank correlation	0.72	—	Descriptive, nine currencies
Cross-currency beta slope	−60	—	Percentage points per beta unit
Cross-currency fitted intercept	−0.75	—	Percentage points at zero beta
Cross-currency fit	$R^2 = 0.77$	—	Descriptive, nine currencies
Beta-sorted spot portfolio	−7.53	0.010	Annualized state difference
Panel state $\times$ beta	−27	0.210	Annualized interaction
Conditional Fama slope, active state	−1.33	0.340	p -value is for the state interaction

Notes: The cross-currency slope, intercept, and R^2 come from regressing the nine bilateral state coefficients on currencies’ sample-average predetermined equity betas. The beta-sorted portfolio uses expanding-window betas. The panel interaction uses monthly predetermined betas. The Fama row is supporting conditional-moment evidence; its episode-level uncertainty is wide.

The beta-sorted portfolio uses a different predetermined ranking variable but the same underlying currency histories. Its spot return is 7.53 percentage points per year lower in the state, with episode-bootstrap $p = 0.010$. The panel interaction between the state and monthly predetermined beta is also negative, about −27 percentage points per beta unit, but has $p = 0.210$. Thus the carry portfolio establishes the return fact, the beta portfolio supports exposure ordering, the nine-currency scatter is descriptive, and the panel interaction is imprecise.

The high-beta distribution also differs from carry’s broad location shift. Its probability of a bottom-decile month rises from 8.5 percent outside the state to 16.3 percent inside ($p = 0.046$), while its variance ratio is 1.05 ($p = 0.80$). The active state is associated with a higher frequency of adverse returns, not a statistically detectable increase in unconditional variance. The cutoff and several distributional tests were examined jointly, so this remains supporting evidence rather than a separate discovery.

The ordering reflects persistent portfolio membership rather than a wholesale signal-month rotation. New Zealand occupies the carry long leg in 85 percent of all months and 91 percent of active-state months; Australia does so in 71 and 73 percent. Japan and Switzerland remain the core funding currencies. Thus the state primarily reprices standing cross-country exposure rather than changing which currencies the sort holds.

The economics is consistent with several established explanations for persistent cur-

rency rankings. Small-country exposure, commodity-export sensitivity, trade-network position, and global intermediary demand can all make high-rate currencies covary more positively with global conditions (Hassan, 2013; Ready et al., 2017; Della Corte et al., 2016; Richmond, 2019; Hassan and Mano, 2019; Gabaix and Maggiori, 2015). Common currency factors and business-cycle exposure provide related asset-pricing interpretations (Lustig and Verdelhan, 2007; Lustig et al., 2011; Lettau et al., 2014; Colacito et al., 2020). The rate-beta rank correlation is 0.72 in this sample; it does not establish that one primitive channel generates both rankings.

5.2 Spot repricing versus predetermined compensation

Seven floating emerging-market currencies provide a directional out-of-construction check because none enters the G10 state definition. All seven bilateral state coefficients are negative. Pooling them gives annualized spot and total-return effects of -14.37 and -13.35 percentage points, with episode-bootstrap p -values of 0.104 and 0.079. The signs and magnitudes are consistent with a broad international state, but the sample overlaps the G10 episodes and is neither an independent signal replication nor a frozen holdout (Bansal and Dahlquist, 2000; Ilzetzki et al., 2019).

The combined G10-plus-emerging-market carry portfolio isolates the role of compensation. On the common 1996–2026 sample, active-state spot returns are almost identical for G10 carry and G10-plus-emerging-market carry: -12.79 and -12.51 percent per year. Total returns differ: -8.67 for G10 and $+2.51$ for G10 plus emerging markets. The implied active-state income legs are approximately 4.12 and 15.02 percent. The state still lowers the broader portfolio’s total return by 9.18 percentage points relative to inactive months, but that estimate is imprecise ($p = 0.235$).

This comparison separates two economic objects that the baseline G10 portfolio alone cannot. Synchronized curves locate a similar international spot repricing across the two carry universes; predetermined interest income determines whether that spot loss produces a negative total mean. The state is therefore more naturally interpreted as a spot-risk state than as a universal negative-total-return rule. Online Appendix C reports the common-sample comparison.

5.3 Supporting conditional moments

Two additional results are economically coherent but statistically weaker. In a conditional Fama regression, the pooled appreciation slope on the lagged interest differential moves from 0.50 outside the state to -1.33 inside it. High-rate currencies therefore depreciate in the active state in point estimates, but the episode-bootstrap p -value for the interaction is 0.340. The result is a descriptive regression image of the carry loss, not precise evidence that uncovered interest parity is restored.

The OECD composite leading indicator declines about 0.6 percent over six months after a state month and remains similar around episode onsets and with lag controls. Industrial production also declines in baseline projections but becomes imprecise with the fullest controls. The leading-indicator response is consistent with the yield-curve literature’s expected-slowdown interpretation (Harvey, 1988; Estrella and Hardouvelis, 1991; Estrella and Mishkin, 1998); the estimates do not establish that synchronized inversions cause activity to fall. Online Appendix C reports the conditional-UIP and local-projection details.

5.4 A compact compensation benchmark

A predictable negative conditional return is not implied mechanically by bad-state exposure. The frictionless Euler condition $\mathbb{E}_t[m_{t+1}rx_{t+1}] = 0$ says that exposure to high-marginal-utility states should command compensation. To state the timing problem compactly, let an adverse event of size $J > 0$ occur at $t + 1$ with conditional probability p_t . Suppose the known carry income can be written as $C_t = \bar{C}_t + \mu J \mathbb{E}_{t-1}[p_t]$, where $\bar{C}_t$ is ordinary carry income and the second term is compensation based on the earlier information set, with $\mu \geq 1$. Then

$$\mathbb{E}_t[rx_{t+1}] = \bar{C}_t + J(\mu \mathbb{E}_{t-1}[p_t] - p_t). \quad (8)$$

A negative conditional mean requires the current perceived loss probability to exceed both previously priced risk and ordinary carry income: $p_t > \mu \mathbb{E}_{t-1}[p_t] + \bar{C}_t/J$. Fresh synchronized inversions can produce that inequality if bond-market information changes before currency compensation fully adjusts. Slow portfolio adjustment or constrained intermediary capacity are possible frictions (Bacchetta and van Wincoop, 2010; Gabaix and Maggiori, 2015); the reduced-form evidence does not identify either one. Online

Appendix B gives the payoff derivation.

5.5 Alternative common-state interpretations

Expected-rate news is one candidate. Synchronized curves can reflect coordinated changes in expected policy paths, but lagging the state establishes chronology rather than exogeneity. Common inflation news is another: international yield-curve components contain substantial inflation variation, and disinflation can synchronize inversions without a global growth shock (Jotikasthira et al., 2015). A common term-premium shock is also possible because an inversion is not a pure expectations measure. The asymmetric ACM result—a robust high-beta effect but an attenuated carry effect after the adjustment—keeps that channel open.

Intermediary constraints can jointly affect bonds and exchange rates. Lagged NFCI and VIX do not span the state, but those variables are incomplete proxies for dealer balance sheets, funding conditions, and global risk-bearing capacity (Brunnermeier et al., 2008; He et al., 2017; Miranda-Agrippino and Rey, 2020). Finally, heterogeneous responses to a broad dollar shock can generate some beta ordering even when the average dollar factor is controlled. The evidence makes a uniform one-for-one dollar account less plausible; it does not eliminate every dollar-based mechanism.

The economic conclusion is comparative. The state predicts losses that are ordered by both interest rates and predetermined equity exposure, and the fitted response is small for near-zero-beta currencies. That pattern supports the interpretation that synchronized inversions contain information about a shared international risk rather than only separate national disturbances. It does not identify the primitive shock or the friction that prices it.

6 Robustness and Limits

The robustness evidence answers three questions. The median, lower quantiles, and bottom-decile deletion show that the return gap extends beyond a few crisis observations. Raw inversion breadth, the U.S. curve, the continuous G10 slope, NFCI, and VIX do not reproduce the result in the reported specifications. Carry coefficients remain negative in both calendar halves and within roughly one-third of the full-sample estimate when any one episode is removed.

An exploratory public-data screen of all two- and three-country delivered-easing combinations produces one family-adjusted CHF–GBP lead, but it weakens under stricter event-count rules and after omitting October 2008. Online Appendix E therefore treats it as a sensor-versus-risk-bearer hypothesis, not a country-driver result.

The exact-one versus exact-two contrast isolates synchronization within the sample. The high-beta portfolio and country gradient support exposure ordering, although the panel interaction is imprecise. Replacing 10Y–2Y slopes with 10Y–3M slopes or removing a U.S. term-premium component attenuates the carry estimate but preserves a statistically detectable high-beta loss. Exposure ordering is less sensitive to these perturbations than the baseline carry magnitude. Conditional UIP and industrial production are imprecise; emerging-market breadth and the downcurve split are exploratory.

Specification selection remains the main uncertainty. The tenor, two-curve threshold, and release rule were assessed on the same return history. A deterministic real-time algorithm can overfit without fitted coefficients. Neighboring-rule grids show whether the baseline is isolated, but cannot exhaust discretion over tenors, cutoffs, countries, freshness, persistence, portfolio size, and timing.

Fifteen episodes limit independent common-state variation. Money-market accounting is not an executable forward strategy after basis, spreads, and timing costs. Predetermined equity beta discriminates among patterns but does not instrument for global growth. U.S. ACM components cannot decompose foreign term premia. Month-end dating prevents mechanical look-ahead but does not audit asynchronous closes, stale quotes, or historical vintages.

The result is a large in-sample predictive association ordered by rates and predetermined equity exposure. It does not establish causality or deployable performance. A frozen sample and feasible forward returns provide the next tests.

7 Conclusion

Synchronized fresh inversions across non-U.S. G10 yield curves predict a distinct currency-return state. Annualized carry-trade spot returns are 12.9 percentage points lower when the lagged state is active, and total carry returns fall from 6.5 to -4.9 percent. The known interest differential remains positive; exchange-rate losses overwhelm it. Median, quantile, and trimmed estimates show that the result is broader than a few crash months.

The cross-section supplies the economic interpretation. High-rate currencies lose relative to low-rate funding currencies, and bilateral losses become larger as predetermined equity-market beta rises. Persistent portfolio membership, a 0.72 alignment between rate and beta rankings, and a beta-sorted portfolio loss show that the state reprices a standing currency characteristic rather than merely rotating the carry portfolio. Directionally similar emerging-market responses extend the pattern beyond currencies used to construct the state, although they are not an independent replication.

This ordering supports the interpretation that synchronized inversions contain information about a shared international risk rather than only separate national disturbances. The evidence does not identify whether that information concerns global growth, coordinated policy, inflation, term premia, intermediary capacity, or a mixture. Slowly adjusting compensation is one coherent explanation, not a recovered structural mechanism.

The estimates use fifteen historical episodes and have not been evaluated in a frozen out-of-sample window. A prospective test should fix the state algorithm and data-vintage protocol before the next synchronized episode occurs.

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Online Appendix to When Yield Curves Invert Together: Predicting Currency Carry Losses

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Abstract

This appendix gives the complete signal algorithm and timing conventions, derives the accounting and candidate compensation mechanism, reports supplementary estimates, and documents reproducible public-data checks, including an exploratory country-combination screen. It separates exact construction details from empirical robustness and from evidence that bears only indirectly on the mechanism.

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A Signal construction and timing

This appendix gives the complete deterministic state transition used in the paper. It fixes the information set and separates fresh entry from confirmed release. It does not establish that the tenor, breadth threshold, and release length were selected independently of the return history.

A.1 Universe and information time

The nine non-dollar G10 currencies are the Australian dollar, Canadian dollar, Swiss franc, euro (Deutsche mark before 1999), British pound, Japanese yen, Norwegian krone, New Zealand dollar, and Swedish krona. The U.S. curve is excluded from the baseline count. For country i , define

$$q_{i,t} = y_{i,t}^{(10Y)} - y_{i,t}^{(2Y)}, \quad I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}.$$

All yields are end-of-month observations. A state measured at the end of month t may condition only returns realized during month $t + 1$.

A.2 Curve-level transition rule

Each curve carries a live indicator $L_{i,t}$, a consecutive-steepening counter $k_{i,t}$, and an eligibility flag. A fresh inversion is a crossing from $q_{i,t-1} \geq 0$ to $q_{i,t} < 0$. The update order is entry, steepening-counter update for curves already live, release after the second consecutive increase, and finally cross-country aggregation.

Table 1: Curve-level state transition

Prior condition	Observation at t	End-of-month action
Eligible and not live	$q_{i,t-1} \geq 0, q_{i,t} < 0$	Set $L_{i,t} = 1$; initialize $k_{i,t} = 0$
Live	$q_{i,t} > q_{i,t-1}$	Increase $k_{i,t}$ by one
Live	$q_{i,t} \leq q_{i,t-1}$	Reset $k_{i,t} = 0$
Live; counter reaches two	Any current slope sign	Set $L_{i,t} = 0$ after the second increase
Released but still inverted	$q_{i,t} < 0$	Remain inactive and ineligible
Released and non-inverted	$q_{i,t} \geq 0$	Become eligible for a future fresh crossing
Any state	Current or lagged slope missing	Create neither a crossing nor a confirmed-steepening update

A curve can therefore be released while still inverted. It cannot re-enter until it first becomes nonnegative and then crosses below zero again. An interrupted steepening sequence resets the counter. These rules make the state depend on the path of slopes, not only on their current signs.

A.3 Cross-country aggregation and raw benchmark

After updating all curves, form

$$N_t = \sum_{i=1}^9 L_{i,t}, \quad S_t = \mathbf{1}\{N_t \geq 2\}.$$

The baseline state S_t is active in 92 of 458 return months and forms fifteen contiguous episodes. The corresponding current-inversion benchmark is

$$R_t = \mathbf{1}\left\{\sum_{i=1}^9 L_{i,t} \geq 2\right\}.$$

The return in month $t + 1$ is paired with S_t or, in the benchmark, R_t . The raw state asks whether fresh entry and confirmed release add information beyond the current sign of the curves.

Table 2: Incidence of the synchronized-inversion state across eras

Era	Sample months	Active months	Active share (%)
1988–94	84	33	39.3
1995–2000	72	19	26.4
2001–07	84	17	20.2
2008–09	24	6	25.0
2010–19	120	3	2.5
2020–21	24	3	12.5
2022–26	50	11	22.0

Notes: Counts reproduce the original era accounting. They describe when the state occurs, not whether its conditional-return coefficient differs across eras.

A.4 Initialization and audit outputs

At the sample start, every curve is initialized as inactive. A curve can enter only after both sides of a fresh crossing are observed. Missing values neither trigger entry nor count toward release. The recursion produces, for every country-month, the slope, raw inversion, fresh-entry flag, live status, steepening counter, release flag, and eligibility status. It then produces N_t , S_t , and episode start, end, and duration. These fields are the minimum audit trail needed to distinguish information month from return month and to reproduce the fifteen-episode count.

B Accounting and mechanism details

This appendix separates return accounting, an exposure-ordering device, and a candidate compensation friction. The first is definitional. The latter two organize the evidence but are not structurally identified by the predictive regressions.

B.1 Currency and portfolio returns

Let $s_{i,t}$ be the log dollar price of currency i and let $d_{i,t} = (i_{i,t} - i_{US,t})/12$. The paper defines the one-month return proxy and carry portfolio as

$$\begin{aligned} rx_{i,t+1} &= d_{i,t} + \Delta s_{i,t+1}, \\ rx_{t+1}^{\text{carry}} &= \frac{1}{3} \sum_{i \in \mathcal{H}_t} rx_{i,t+1} - \frac{1}{3} \sum_{i \in \mathcal{F}_t} rx_{i,t+1} \equiv C_t + X_{t+1}. \end{aligned} \quad (1)$$

The high-rate set is $\mathcal{H}_t$ and the funding set is $\mathcal{F}_t$. The construction makes $C_t > 0$ in every sample month, so the portfolio loses only when $X_{t+1} < -C_t$. Empirically, C_t averages 3.9 percent per year and has a minimum of 0.7 percent. The active-state spot-return difference is -12.85 percentage points and the total-return difference -11.31 , so known income absorbs only part of the currency loss.

To organize the cross-section, write

$$\Delta s_{i,t+1} = -\lambda_i g_{t+1} + \eta_{i,t+1}, \quad (2)$$

where $g_{t+1} > 0$ is an adverse component shared across currencies, λ_i is a latent loading, and $\eta_{i,t+1}$ is idiosyncratic. A diversified long-short portfolio partly reduces idiosyncratic components; it retains the shared component when average λ_i differs between its two legs. The expanding-window equity beta $\hat{\beta}_{i,t}$ is a predetermined proxy for λ_i , not the same primitive parameter.

B.2 Yield slopes and synchronized information

A long-minus-short yield spread combines differences in expected short-rate averages with differences in maturity-specific term premia (Campbell and Shiller, 1991; Hamilton and Kim, 2002). Consequently, one inversion is consistent with expected easing but does not reveal whether the news is domestic, global, expectations-driven, or a term-premium movement. Synchronization raises the plausibility of shared information because several markets cross a salient boundary, but it does not identify the shared shock.

The exact-one versus exact-two estimates give this idea empirical content. Exactly one live inversion is associated with carry spot and total coefficients of 8.33 and 8.78 percentage points per year. At exactly two, the coefficients are -15.72 and -15.13 . The sign reversal indicates nonlinearity in cross-country breadth within the sample. It does not establish two as an optimal population threshold.

B.3 Delayed compensation as a candidate friction

Let A_{t+1} indicate an adverse event that reduces carry payoff by $J > 0$. Write

$$rx_{t+1}^{\text{carry}} = C_t - JA_{t+1} + \xi_{t+1}, \quad \mathbb{P}_t(A_{t+1} = 1) = p_t, \quad \mathbb{E}_t[\xi_{t+1}] = 0. \quad (3)$$

Then $\mathbb{E}_t[r_{t+1}^{\text{carry}}] = C_t - p_t J$. For a tradable excess return in a frictionless benchmark, the Euler condition $\mathbb{E}_t[m_{t+1} r_{t+1}^{\text{carry}}] = 0$ requires compensation when losses occur in high-marginal-utility states. Publicly visible risk therefore does not automatically imply a negative conditional mean.

The candidate friction places an information lag in compensation. Suppose

$$C_t = \bar{C}_t + \mu J \mathbb{E}_{t-1}[p_t], \quad \mu \geq 1, \quad (4)$$

where $\bar{C}_t$ is ordinary carry income and the second component prices the current-period hazard using the earlier information set. Combining equations (3) and (4) gives

$$\mathbb{E}_t[r_{t+1}^{\text{carry}}] = \bar{C}_t + J \{ \mu \mathbb{E}_{t-1}[p_t] - p_t \}. \quad (5)$$

Hence

$$\mathbb{E}_t[r_{t+1}^{\text{carry}}] < 0 \iff p_t > \mu \mathbb{E}_{t-1}[p_t] + \frac{\bar{C}_t}{J}. \quad (6)$$

The inequality is the model's economic content. A rise in perceived risk is insufficient: the current expected loss must exceed both previously embedded event compensation and ordinary carry income. Fresh synchronized inversions may mark such a belief innovation, but the paper does not observe p_t or identify the adjustment friction.

B.4 Conditional uncovered interest parity

The supporting Fama regression estimates the appreciation slope separately by state:

$$\Delta s_{i,t+1} = a_i^{(s)} + \phi_s d_{i,t} + u_{i,t+1}, \quad S_t = s, \quad s \in \{0, 1\}. \quad (7)$$

Under the paper's quote convention, uncovered interest parity implies $\phi_s = -1$. The pooled estimate is 0.50 outside the state and -1.33 inside it. The point estimates say that high-rate currencies depreciate in the state in which carry loses. The state interaction has a Newey–West t -statistic of -2.39 but an episode-bootstrap p -value of 0.340. The pooled panel contains many currency-months, whereas the common state changes across only fifteen episodes; the episode result is therefore the appropriate warning against treating -1.33 as a precise structural slope.

B.5 Exploratory curve-shape compensation

Define the cross-country average yield at maturity m as $\bar{y}_t^{(m)} = 9^{-1} \sum_{i=1}^9 y_{i,t}^{(m)}$. An active-state month is classified as downcurve when

$$\bar{y}_t^{(3M)} > \bar{y}_t^{(2Y)} > \bar{y}_t^{(10Y)}. \quad (8)$$

There are 52 downcurve months and 35 overlap the synchronized state. In those 35 months, carry income averages 6.48 percent per year and the spot leg -0.96 ; in the other 57 active-state

months, income averages 4.31 and the spot leg -15.55 . Total carry returns average 5.51 percent in downcurve-state months and -11.24 percent in the remaining state months.

This split is economically suggestive: high, fully downward-sloping curves can mark inflation normalization with unusually large carry income, whereas other synchronized inversions can look more like uncompensated stress. The classification was developed after examining apparent false alarms and is concentrated in 1988–91 and 2023–24, plus one month in 2007. Rate level, inflation, and curve shape move together in those periods, so the heterogeneity is exploratory and does not establish a feasible trading rule.

C Supplementary evidence and robustness

All exercises in this appendix use the same historical sample on which the state was assessed. They measure sensitivity and economic breadth; they do not create a frozen out-of-sample test.

C.1 Distribution and episode influence

Removing each portfolio’s full-sample bottom-decile months leaves negative carry spot and total state effects. Because the deletion conditions on the realized outcome, it diagnoses tail concentration rather than estimating a preferred conditional mean.

Table 3: State effects after removing bottom-decile months

Portfolio component	Remaining signal months	Difference (%/yr)	Episode wild p
Carry spot	77	-7.24	0.006
Carry total	77	-5.50	0.024
High-beta spot	77	-3.72	0.084
High-beta total	78	-3.68	0.060

Notes: Each regression drops that return series’ full-sample bottom-decile months from both states. The exercise asks whether a few crash observations mechanically generate the conditional-mean result; it does not address selection among alternative signal definitions. Wild-bootstrap p -values cluster by IYC episode.

The median and every reported lower quantile also move left. For carry spot returns, the monthly median changes from 0.35 percent outside the state to -0.76 percent inside, the 25th percentile from -0.93 to -1.98 , and the 10th percentile from -2.35 to -3.94 . This evidence is more informative about a broad distributional shift than the fact that all five worst months occur under the lagged state.

Leave-one-episode-out carry estimates remain within roughly one-third of the full-sample coefficient. The high-beta effect remains statistically detectable under every one-episode deletion, with a largest reported bootstrap p -value of 0.006. These checks reduce dependence on a single episode but do not overcome the limited number of episodes.

C.2 Calendar stability

The carry-spot coefficient is negative in both calendar halves. The high-beta result is less stable and is concentrated after 2005. The split is descriptive because the rule was not frozen on the first half before evaluation on the second.

Table 4: The headline estimates in two halves of the sample

Series	1988:01–2004:12			2005:01–2026:02		
	b	$t_{episode}$	wild p	b	$t_{episode}$	wild p
Carry spot	-11.41	-2.54	0.037	-15.85	-1.99	0.010
Carry total	-10.20	-2.04	0.052	-14.76	-1.84	0.047
High-beta spot	-4.01	-1.12	0.287	-11.91	-4.07	0.010
High-beta total	-3.03	-0.80	0.448	-11.54	-3.85	0.011
Months	204			254		
Signal months	54			38		
Episodes	7			8		

Notes: Each column reports the state regression $y_t = a + b \mathbf{1}\{\text{IYC}\}_{t-1} + e_t$ estimated separately by calendar half. Coefficients are annualized percentage points; $t_{episode}$ clusters by IYC episode, and wild-bootstrap p -values use $B = 9,999$. This is a stability check, not a frozen out-of-sample test.

C.3 State-definition sensitivity

The original grids vary breadth, release confirmation, and the short maturity. Exactly one live inversion has positive carry coefficients, while exactly two produces the largest negative coefficients. At-least-three and at-least-four rules have fewer active months and later coverage. Releasing after one steepening month is noisier; requiring three retains the state longer. Rebuilding the state with a 10Y–3M rather than 10Y–2Y slope preserves the broad adverse ordering but changes episode timing.

The raw current-inversion state is active in 165 months, compared with 92 for the live state. It captures fewer of the worst carry months and has smaller return coefficients. The live state can retain losses that arrive after curves begin to re-steepen, but its additional tail coverage is partly mechanical because the historical crash chronology helped motivate persistence.

A reciprocal-age version of the current-inversion count separates two timing margins. It has a high-beta spot coefficient of -14.87 ($p = 0.001$), larger than the baseline live state’s -7.53 , but a smaller carry coefficient of -10.43 ($p = 0.040$) than the baseline’s -12.85 . Under the full control set, the age-weighted high-beta coefficient remains -13.82 ($p = 0.008$), whereas its carry coefficient attenuates to -7.55 ($p = 0.084$). These in-sample comparisons suggest that exposure repricing is concentrated near fresh information, while the path-dependent release rule is more important for carry losses that continue into re-steepening.

Table 5: Measurement perturbations and exposure ordering

State construction	Carry spot	Wild p	High-beta spot	Wild p
10Y–2Y baseline	−12.85	0.001	−7.53	0.010
10Y–3M slope	−4.68	0.146	−5.25	0.011
U.S.-term-premium-adjusted 10Y–2Y	−5.42	0.107	−6.17	0.004

Notes: Coefficients are annualized percentage-point active-state differences. The 10Y–3M row rebuilds every curve state using a different short maturity. The term-premium row removes each national slope’s fitted exposure to the U.S. ACM 10Y-minus-2Y slope premium before rebuilding the state. Both perturbations attenuate carry while retaining the beta-sorted exposure result.

Table 6: State-definition multiverse: reported and untested dimensions

Dimension	Reported neighborhood	Remaining uncertainty
Breadth	At least 1–4; exactly 1–2	Frequency-matched and country-combination rules
Release	One-, two-, and three-month confirmation	Other persistence and duration-matched placebo rules
Tenor	10Y–2Y baseline; 10Y–3M check	Other maturities and inversion cutoffs
Freshness	Fresh crossing versus current inversion	Alternative age decay and entry delay
Geography	Non-U.S. G10 baseline; U.S.-inclusion checks	Other developed-market sets and country combinations
Portfolio	Three-by-three carry and beta sorts	Other leg sizes, volatility scaling, feasible forwards
Timing	Month-end recursion and one extra lag	Vendor timestamps, asynchronous closes, vintage data

C.4 Exposure breadth and composition

The carry and beta portfolios hold persistent characteristics rather than changing membership only when the state turns on. New Zealand occupies the high-rate leg in 85 percent of all months and 91 percent of state months; Australia in 71 and 73 percent. Japan and Switzerland remain the principal funding currencies. This makes the state result more naturally a repricing of standing exposure than a mechanical change in portfolio composition.

The seven floating emerging-market currencies are not used to construct the G10 state. All seven bilateral state coefficients are negative; pooled spot and total effects are −14.37 and −13.35 percentage points per year, with episode-bootstrap p -values of 0.104 and 0.079. This is directional breadth, not an independent replication, because the same global dates define the state and the emerging-market sample is shorter.

The common-sample comparison separates international spot repricing from compensation.

Table 7: Spot repricing and compensation across currency universes, 1996–2026

Universe	Return leg	Inactive mean	Active mean	State difference	Episode wild p
G10	Spot	2.96	−12.79	−15.75	< 0.001
G10	Total	6.22	−8.67	−14.89	0.003
G10+EM	Spot	0.46	−12.51	−12.97	0.176
G10+EM	Total	11.68	2.51	−9.18	0.235

Notes: Annualized percentages over the common 1996:01–2026:02 sample. The G10+EM universe adds seven floating emerging-market currencies and applies the same three-by-three rate sort. Active-state income implied by total minus spot is approximately 4.12 percent for G10 and 15.02 percent for G10+EM. The nearly equal spot means but different total means show how predetermined carry income changes the payoff from similar currency repricing.

All reported exposure tests use expanding S&P 500 betas. An oil-beta sort adds little independent variation because its return is largely spanned by the carry and equity-beta portfolios.

C.5 Conditional UIP and macro timing

Appendix B gives the conditional Fama specification. The pooled slope is 0.50 outside the state and −1.33 inside, but the state interaction has episode-bootstrap $p = 0.340$. The point estimate is consistent with high-rate currencies depreciating during the state, while the interaction remains imprecise at the episode level.

The leading indicator declines by about 0.6 percent over six months, with similar magnitudes around episode onsets and under lag controls. Industrial production declines about 1.8 percent at six months in the baseline projection but becomes imprecise with the fullest controls. The leading-indicator response is persistent across these specifications; the industrial-production response is not.

C.6 Benchmarks, controls, and term premia

The U.S. inversion, average G10 slope, and its change do not span the synchronized state. Lagged NFCI and VIX controls also leave the carry coefficient near its baseline magnitude. Contemporaneous dollar and FX-volatility controls show that the realized loss is not simply the average dollar move or the measured volatility innovation, but they are not predictive controls.

The term-premium adjustment is asymmetric. After removing fitted exposure to the U.S. ACM 10Y-minus-2Y term-premium slope, 83 percent of national inversions remain. The high-beta coefficient is −6.2 percentage points with $p = 0.004$. The carry coefficient is about −5.4 with $p = 0.11$, roughly half the baseline magnitude. A common U.S. term-premium component therefore does not eliminate exposure ordering, but may explain part of the carry association. Comparable foreign-country term-premium decompositions are unavailable.

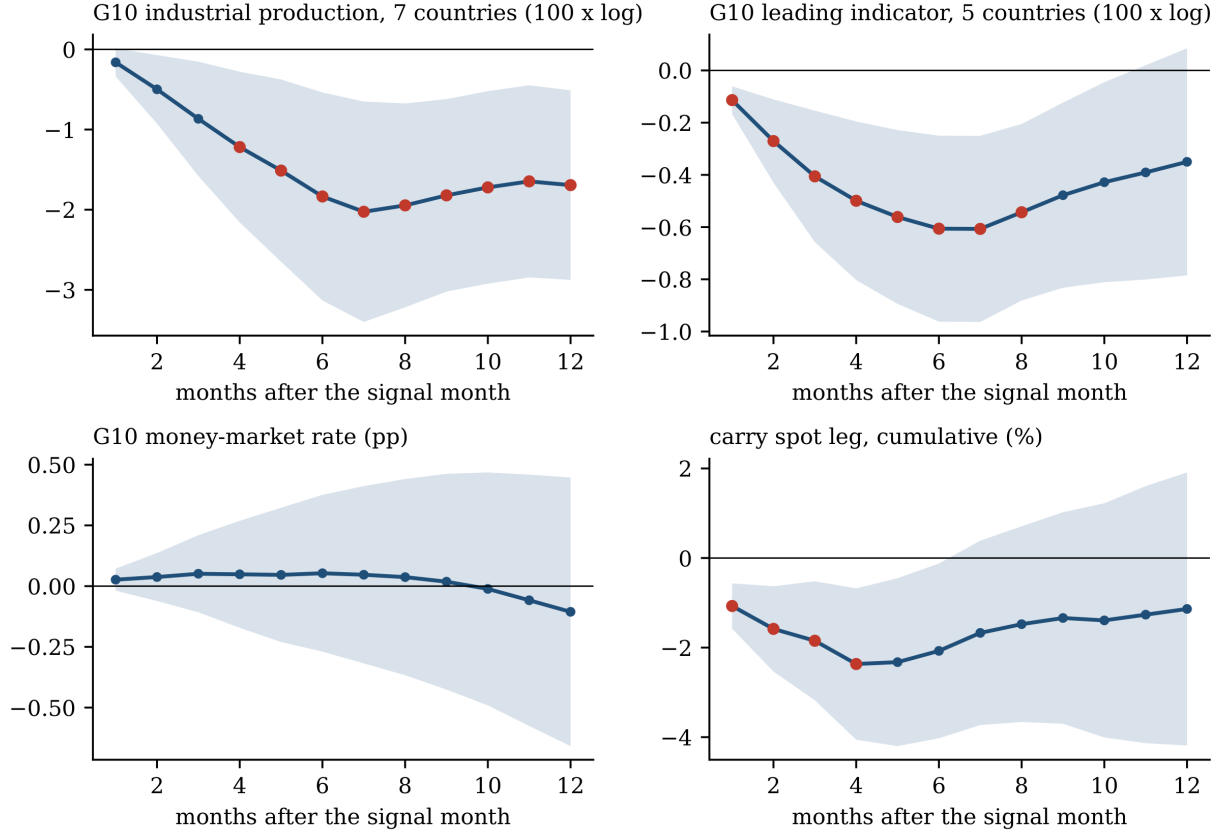


Figure 1: Outcomes following a synchronized-inversion state. The panels show local projections of cumulative changes in G10 industrial production, the OECD composite leading indicator, the average G10 money-market rate, and the carry spot leg. Bands are 90-percent Newey–West intervals; markers denote episode-bootstrap $p < 0.10$. Overlapping horizons and persistent states make these supporting temporal associations rather than causal responses.

C.7 Interpretive boundary

The conditional return gap, distributional shift, and exposure ordering are statistically detectable in the reported specifications. The conditional-UIP interaction and industrial-production response are imprecise; emerging-market breadth and the downcurve split are exploratory. Feasible forward returns, complete data-vintage auditing, a comprehensive state multiverse, and a frozen future sample remain untested.

D Public-data mechanism checks

This appendix uses public data to examine outcomes around synchronized delivered policy easing. The event is intentionally later than the yield-curve state: inversions can encode expected easing, whereas policy-rate cuts record policymakers' actions. The exercise tests selected mechanism implications but, because it observes policy delivery rather than the predictive state, it does not validate the inversion classifier.

D.1 Event construction

The sample is January 1988 through December 2025. Let $r_{i,t}^{pol}$ be country i 's BIS end-of-period policy rate and define

$$K_{i,t} = \mathbf{1}\{\Delta r_{i,t}^{pol} \leq -0.10\}, \quad B_t = \sum_i K_{i,t}, \quad (9)$$

where rates and the 10-basis-point cutoff are in percentage points. Let $D_t = 1$ when $B_t \geq 3$ and at least six country rates are observed. The onset indicator is

$$O_t = D_t \prod_{\ell=1}^3 (1 - D_{t-\ell}). \quad (10)$$

Thus an onset requires at least three cutting countries after three months without another synchronized-cut month. The rule produces fifteen onsets. These fifteen public easing onsets are not the paper's fifteen inversion episodes; the identical counts are coincidental.

The public currency outcome is a spot-only proxy. BIS exchange rates are signed so that a positive return is foreign-currency appreciation against the dollar. At $t-1$, currencies are ranked by their policy-rate differentials against the United States; the proxy is long the three highest-ranked and short the three lowest-ranked. The rank is predetermined, but O_t is observed during month t , so month- t returns are contemporaneous event associations rather than feasible predictions made before easing.

Additional outcomes are CFTC non-commercial net positions scaled by open interest, the New York Fed ACM 10-year expected-rate and term-premium components (Adrian et al., 2013), the OECD G7 composite leading indicator, NFCI, and VIX. CFTC coverage is limited to seven directly matchable contracts and uses a documented Deutsche-mark/euro convention. ACM is a U.S.

diagnostic, not a decomposition of the nine foreign slopes. The macro and financial-condition series are current vintage.

D.2 Event estimands and inference

The code uses two estimands. For a level outcome Y , the horizon- h event response is

$$\hat{\theta}_h^{\text{change}} = \frac{1}{n_h} \sum_{j \in \mathcal{J}_h} (Y_{\tau_j+h} - Y_{\tau_j-1}). \quad (11)$$

For a monthly return flow R , it is

$$\hat{\theta}_h^{\text{return}} = \frac{1}{n_h} \sum_{j \in \mathcal{J}_h} \sum_{\ell=0}^h R_{\tau_j+\ell}, \quad (12)$$

where τ_j indexes valid onset dates. Ninety-percent intervals resample onset episodes; leave-one-onset ranges diagnose influence. The reference distribution enumerates circular shifts of the complete onset sequence, retains shifts with the observed number of valid event outcomes, includes the observed assignment, and doubles the smaller inclusive-tail rank. With irregular missingness, the retained shifts need not form a transformation group; the resulting values are conditional finite-rotation references, not exact causal randomization tests.

Six declared primary outcomes receive Holm adjustment: the shadow-carry spot return over months 0–1; the CFTC positioning change through month 3; ACM expected-rate and term-premium changes through month 3; the CLI change through month 6; and the NFCI change through month 3. The declaration is recorded in the analysis configuration and run manifest, but not in an immutable pre-analysis registration. VIX and other horizons are descriptive.

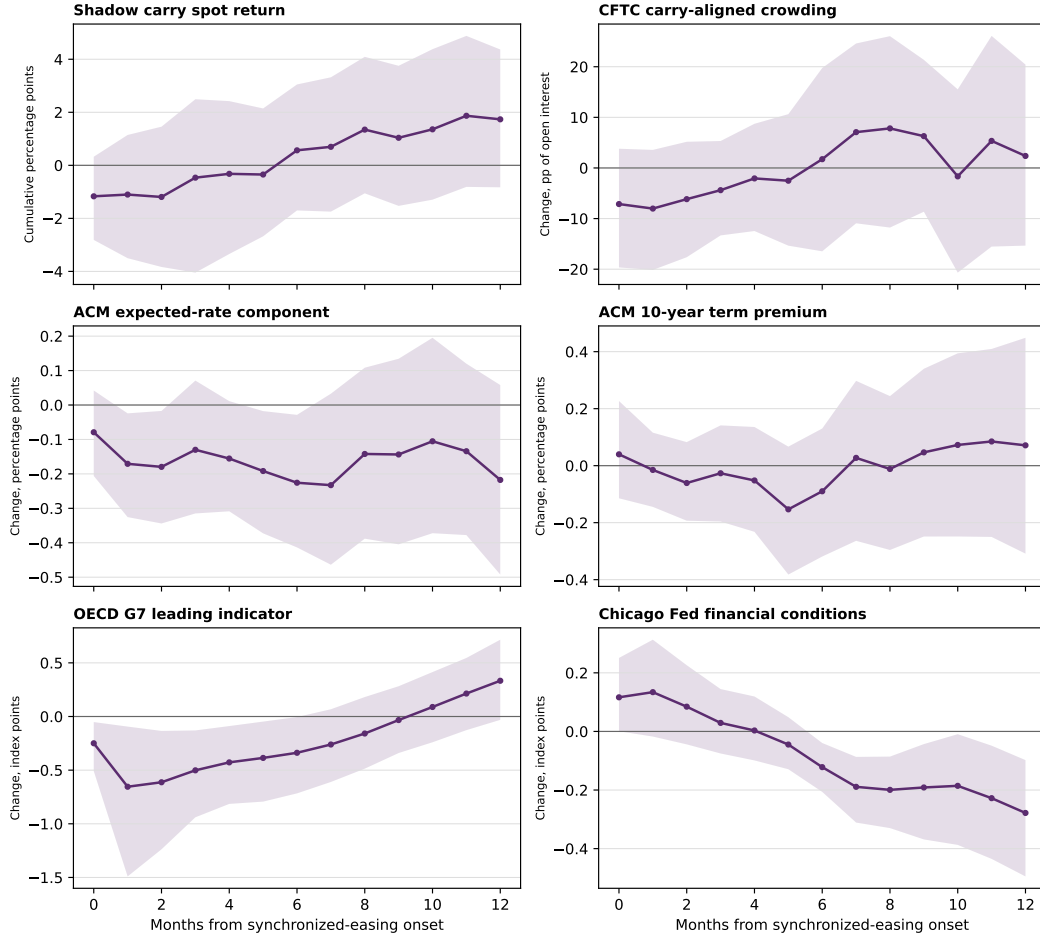
The signs mostly fit a broad stress-and-easing chronology: the shadow carry proxy depreciates, carry-aligned positions decline, the expected-rate component falls, and the leading indicator weakens. Estimates are imprecise, and no primary outcome survives the Holm adjustment. The U.S. term-premium estimate is nearly flat at three months. These results are directional mechanism evidence around policy delivery, not evidence that the inversion state predicts them.

Table 8: Outcomes around synchronized delivered-easing onsets

Outcome	Estimate	90% CI	RI p	Holm p	Rotations
Policy-ranked shadow carry spot return, months 0–1	-1.10	[-3.55, 1.18]	0.195	0.975	441
Carry-aligned CFTC non-commercial net share, change through month 3	-4.39	[-13.53, 5.40]	0.741	1.000	81
ACM 10-year expected-rate component, change through month 3	-0.13	[-0.31, 0.07]	0.202	0.975	396
ACM 10-year term premium, change through month 3	-0.03	[-0.19, 0.14]	0.980	1.000	396
OECD G7 CLI, change through month 6	-0.34	[-0.72, 0.00]	0.159	0.952	353
Chicago Fed NFCI, change through month 3	0.03	[-0.08, 0.15]	0.520	1.000	396
VIX, change through month 1	1.89	[-2.30, 5.81]	0.400	—	20

Notes: Events are synchronized delivered-easing onsets constructed from public BIS policy rates. They are a downstream policy-rate state, distinct from the paper’s predictive IYC state. RI p -values enumerate every circular rotation, condition on the observed number of valid episode outcomes, include the observed assignment, and double the smaller inclusive tail rank. Rotations reports the resulting finite reference count. The CFTC reference is small because missing contract coverage makes most rotations change the valid event count; its RI result is correspondingly coarse. With irregular missingness, the same-count rotations need not form a transformation group, so these are conditional finite-rotation references rather than exact causal randomization tests. Intervals resample onset episodes. Holm p -values adjust the six prespecified primary outcomes. Currency returns and CFTC positioning are percentage points, with CFTC positions measured as a share of open interest; ACM components are percentage points; CLI, NFCI, and VIX are index points.

Public proxies around synchronized delivered easing



Means across onset episodes; bands are 90% episode-bootstrap intervals. The delivered-easing state is downstream of, and distinct from, the predictive IYC state.

Figure 2: Outcomes around synchronized delivered-easing onsets. Event time zero is a month in which at least three policy rates fall by at least 10 basis points after a three-month quiet period. Panels show the code-defined cumulative return or endpoint-change estimands, with 90-percent episode-bootstrap intervals.

D.3 Sensitivity and interpretation

The shadow-carry response is recomputed using two- and four-country event thresholds and after omitting each event currency. Every threshold recomputes its own quiet period, so onset counts need not be monotone in the required number of cutting countries.

The exercise has four binding limits. Delivered easing is endogenous to prior information and occurs downstream of the paper’s predictor. Policy-rate ranks are not money-market or forward rates, and spot returns omit carry income, basis, transaction costs, and implementation lags. CFTC, ACM, and current-vintage macro series are imperfect proxies for the intended objects. Finally, fifteen onsets—fewer for some outcomes—leave low episode-level power. The public results are therefore useful as a reproducible challenge to mechanism implications, not as a replication of

Table 9: Public-proxy sensitivity: shadow carry spot return

Construction	Estimate	RI p	Events	Rotations
At least three countries easing	-1.10	0.195	15	441
At least two countries easing	-1.24	0.060	22	434
At least four countries easing	-0.46	0.635	18	438
Exclude AUD from event rule	-0.50	0.556	17	439
Exclude CAD from event rule	0.36	0.562	18	438
Exclude CHF from event rule	-1.21	0.150	15	441
Exclude EUR from event rule	-1.00	0.214	16	440
Exclude GBP from event rule	-0.81	0.321	20	436
Exclude JPY from event rule	-0.27	0.787	19	437
Exclude NOK from event rule	-1.65	0.063	13	443
Exclude NZD from event rule	-0.53	0.580	15	441
Exclude SEK from event rule	-0.47	0.606	17	439

Notes: Each threshold definition recomputes its own three-month quiet window. Onset counts therefore need not be monotone in the number of countries required to cut. Leave-one-currency rows omit that currency from the event rule.

the headline state.

E Which countries matter in the public proxy?

This appendix asks whether particular country pairs or triples dominate the public delivered-easing association. It does not observe the original nine-country yield-slope panel and therefore cannot identify which curves created the paper’s fifteen inversion episodes. The analysis uses the same BIS policy-rate and exchange-rate data as Appendix D.

E.1 Combination screen

All 36 currency pairs and 84 triples are enumerated. For a given combination, an onset occurs when every member cuts its policy rate by at least 10 basis points after three months without that same joint cut. The outcome is the public shadow-carry spot return accumulated over event months zero and one. The ranking of all nine currencies uses lagged policy-rate differentials; the tested country combination defines the event, not the portfolio.

Inference is reported only for combinations with at least six valid events. Every eligible event sequence is shifted through every circular calendar rotation. Raw two-sided rotation p -values are supplemented with a common-rotation maximum absolute standardized statistic that controls family-wise error across every eligible pair and triple. Episode-resampling intervals, leave-one-event means, and alternative minimum-event rules diagnose small-sample fragility.

Table 10: Country combinations in the public delivered-easing proxy

Combination	Events	Mean $[0, 1]$	Raw p	Family-adjusted p
CHF–GBP	6	−4.63	0.009	0.042
AUD–EUR	6	−4.06	0.009	0.088
GBP–SEK	9	−3.28	0.018	0.152
CHF–SEK	7	−2.74	0.070	0.549
CHF–NOK	8	−2.62	0.057	0.538
CAD–NOK–NZD	7	−4.29	0.013	0.055
NOK–NZD–SEK	6	−3.62	0.026	0.279
CAD–GBP–NOK	6	−3.58	0.022	0.277
GBP–NZD–SEK	8	−3.52	0.018	0.143
AUD–NOK–NZD	6	−3.11	0.044	0.440

Notes: The table reports the five most negative eligible pairs and five most negative eligible triples. An event requires every listed country to cut its BIS policy rate by at least 10 basis points after three months without the same joint cut. The outcome is the cumulative policy-ranked shadow-carry spot return in event months 0 and 1, in percentage points. Combinations require at least six valid events. Raw two-sided p -values use every circular timing rotation. The final column controls family-wise error across all 42 eligible pairs and triples with a common-rotation maximum absolute standardized statistic. This is a delivered-easing proxy, not a replication of the yield-curve state.

Of 120 combinations, 42 have at least six events. CHF–GBP is the only 5-percent family-wide result: six joint-cut onsets have a two-month mean shadow-carry spot return of −4.63 percentage points, raw $p = 0.009$, and adjusted $p = 0.042$. The 90-percent episode-bootstrap interval is $[-9.52, 0.26]$, however, and deleting October 2008 reduces the mean to −2.62. CAD–NOK–NZD is next, with seven events, a −4.29-point mean, and adjusted $p = 0.055$.

The family result depends on admitting sparse combinations. With a six-event minimum, 42 combinations are eligible and one clears 5 percent. With minimums of eight or ten events, 25 and 15 combinations remain, no adjusted result clears 5 percent, and the smallest adjusted p -values are 0.105 and 0.224. The evidence does not establish a stable country bloc.

E.2 Sensor countries versus loss-bearing currencies

The public lead does not match the baseline loss-bearing cross-section. CHF and GBP have near-zero or modest equity betas and small, imprecise state-conditioned currency losses. AUD and NZD bear the larger losses. The own-curve horse race also shows that a country's own curve generally does not forecast its currency after conditioning on the synchronized state.

This contrast suggests a testable distinction: countries whose bond or policy states reveal a shared episode need not be the currencies that bear its losses. Joint Swiss–U.K. easing may function as a calendar sensor for broad stress, while high-rate, high-beta currencies absorb the carry loss. The distinction is economically consistent with synchronization and exposure ordering, but the present result is only hypothesis-generating because delivered easing is downstream of forward-looking yield-curve inversions.

The direct test requires the nine-country live-state matrix. A fifteen-episode-by-nine-country contributor ledger could enumerate which curves activate each episode, then compare contributor combinations with the subsequent currency-loss cross-section under family-wide inference. The public combination screen specifies that future test but does not decompose the headline result.

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